

ignores the imaginary component in all other uses of this function.

2. 'Loglog(Y)' plots the columns of Y versus their index if Y contains real numbers. If Y contains complex numbers, 'loglog(Y)' and 'loglog(real(Y),imag(Y))' are equivalent. 'Loglog' ignores the imaginary

component in all other uses of this function.

3. 'Loglog(X1,Y1,...)' plots all Xn versus Yn pairs. If only Xn or Yn is a matrix, loglog plots the vector argument versus the rows or columns of the matrix, depending on whether the vector's row or column dimension

matches the matrix.

EFY note. The source codes of this article are available on www.efymag.com website.

Nishant Mittal is a final-semester student of BE in electronics and telecommunication engineering and Dhanalekshmi Menon is an ME student in digital systems, MAEERS, MIT, Pune

IIR.M

```

clc;
clear;
wc=input('Enter the 3db frequency = ');
w0=100*wc;
T=pi/w0;
fs=1/T;
hdb=fs*wc;
hdb=[1 -exp(-T*wc)];
hab=wc;
haa=[1 wc];
F0:(wc/(2000*pi)):(10*wc/(2*pi));

w=2*pi*F;
hd=freqz(hdb,hda,F,fs);
hdnorm=(hd/hd(1));
ha=freqz(hab,haa,w);
figure(1);
loglog(F,20*log10(abs(hdnorm)),'r',F,20
*log10(abs(ha)),'b');
title('Magnitude ');
grid on;
xlabel('log freq -->');
ylabel('log amplitude -->');

figure(2);
semilogx(F,angle(hdnorm),'r',F,angle(ha),
'b');
title('Phase ');
grid on;
xlabel('log freq -->');
ylabel('Amplitude -->');
%Output.
%Enter the 3db frequency = 150

```

RECTANGULAR.M

```

clc;
clear;
N=input('Enter the length of filter = ');
for n=1:1:N
    W(n)=1;
end
k=0;
c=1;
for c=1:1:629
    X(c)=0;
end
for w=-pi:0.01:pi
    for n=1:N
        X(k)=X(k)+(W(n)*exp(-1)*w*(n-1));
    end
    M(k)=abs(X(k));
    A(k)=angle(X(k));
    k=k+1;
end
M=20*log10(M);
w=-pi:0.01:pi;
n=0:N-1;
figure(1);
stem(n,M);
grid on;
title('Filter');
xlabel('samples -->');
ylabel('Amplitude -->');

figure(2);
plot(w,M);
grid on;
title('Magnitude ');
xlabel('samples -->');
ylabel('Amplitude -->');
figure(3);
plot(w,A);
grid on;
title('Angle');
xlabel('samples -->');
ylabel('Amplitude -->');
%Output.
%Enter the length of filter = 31

```

HAMMING.M

```

clc;
clear;

N=input('Enter the length of filter = ');
for n=1:1:N
    W(n)=.54-(.46*cos(2*pi*n/(N-1)));
end
k=0;
c=1;
for c=1:1:629
    X(c)=0;
end
for w=-pi:0.01:pi
    for n=1:N
        X(k)=X(k)+(W(n)*exp(-1)*w*(n-1));
    end
    M(k)=abs(X(k));
    A(k)=angle(X(k));
    k=k+1;
end
M=20*log10(M);
w=-pi:0.01:pi;
n=0:N-1;
figure(1);
stem(n,M);
grid on;
title('Filter');
xlabel('samples -->');
ylabel('Amplitude -->');

figure(2);
plot(w,M);
grid on;
title('Magnitude ');
xlabel('samples -->');
ylabel('Amplitude -->');
figure(3);
plot(w,A);
grid on;
title('Angle');
xlabel('samples -->');
ylabel('Amplitude -->');
%Output.
%Enter the length of filter = 31

```



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